

Readability Enhancement of Displayed Images Under Ambient Light

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Abstract—Image quality in mobile displays is considerably influenced by ambient light. In the daylight condition, images on mobile displays are darkly perceived by the human visual system due to the limited dynamic range of display, which causes loss of luminance and details. In this paper, we propose readability enhancement of displayed images under ambient light by enhancing both luminance and details. We design a weighted optimization framework, which contains the data term for luminance enhancement and the gradient term for detail enhancement. In the data term, we use an ambient light nonlinear intensity-transfer function considering display properties, ambient light, and image contents. In the gradient term, we employ a threshold versus intensity adaptation function based on the degree of ambient light adaptation to compensate for gradient distortions. We solve the optimization framework using a numerical solver and achieve image enhancement in displayed images. Experimental results demonstrate that the proposed method significantly improves readability of mobile displays under ambient light by enhancing luminance and details of images.

Index Terms—Display enhancement, ambient light, human visual system, mobile display, luminance enhancement, detail enhancement.

I. INTRODUCTION

IN RECENT years, smart phones play an important role as an information delivery system. Many display manufacturers concentrate on improving display properties such as contrast ratio, screen size, backlight intensity, and viewing angle to get good visual quality and high color fidelity. However, changes in viewing condition, especially illumination, seriously degrade visual quality of displayed images [1]. Thus, besides the display technology, viewing condition attracts the attention in perceptual-based modeling very recently, such as quality evaluation [2]. One of illumination properties, the color temperature, makes displayed images perceive reddish or bluish. Moreover, we frequently encounter different lights between office and outdoor environments, which degrades contrast, colorfulness, and display gamut size of displayed images by elevating the black level of displays.

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In particular, under daylight condition, displayed images on mobile displays are darkly perceived by HVS and its details are reduced largely. In this work, uniform illumination, i.e. ambient light, is considered as a main viewing condition factor to simplify the illumination problem. Due to the reflected light from display screens and the adaption of HVS when people view them, ambient light causes degradation of displayed images in color appearance and contrast.

A. Related Work

Up to now, many results have been achieved by researchers to deal with the ambient light problem [3]–[19]. First, image enhancement methods [3], [4] were proposed to solve it, but didn't produce good results without considering ambient light. Then, two groups of methods considered ambient light were studied: 1) color appearance enhancement and 2) contrast enhancement.

The one group focuses on the effect of ambient light on the color appearance of displayed images. CIECAM 97s [5] and CIECAM02 [6] were developed by the Commission Internationale de L'Eclairage (CIE). The two color appearance models included a set of perceptual effects to compensate color appearance distortion under different surrounding conditions such as dark, dim, and normal scenes. Katoh et al. proposed a new adaptation model, S_LMS, to compensate color appearance distortion under mixed illumination, i.e. person adapts to mixed color temperature of ambient light and self-luminous displays [7]. Laine proposed varying three parameters of CIECAM 97s color appearance model, i.e. chromatic adaptation parameter D , the surround compensation parameters c and D_c to obtain the same color appearance as the original image under different ambient illuminations [8]. Although color appearance was predicted under different ambient lights, these models were required to determine many viewing condition parameters, and did not consider strong ambient light in daylight.

The other group analyzes the effect of varying ambient lights on luminance or contrast distortion of displayed images. Drago *et al.* [9] proposed a logarithmic power function for luminance enhancement based on many subjective experiments. This method effectively increased the perceived luminance of image, but caused washing-out color artifacts for the enhanced image. Kim focused on backlight intensity control based on ambient light intensity by lux sensor [10]. This method led to the power consumption problem. Monobe *et al.* [11] proposed a method to preserve the same local contrast under ambient light as the original image in the dark environment.

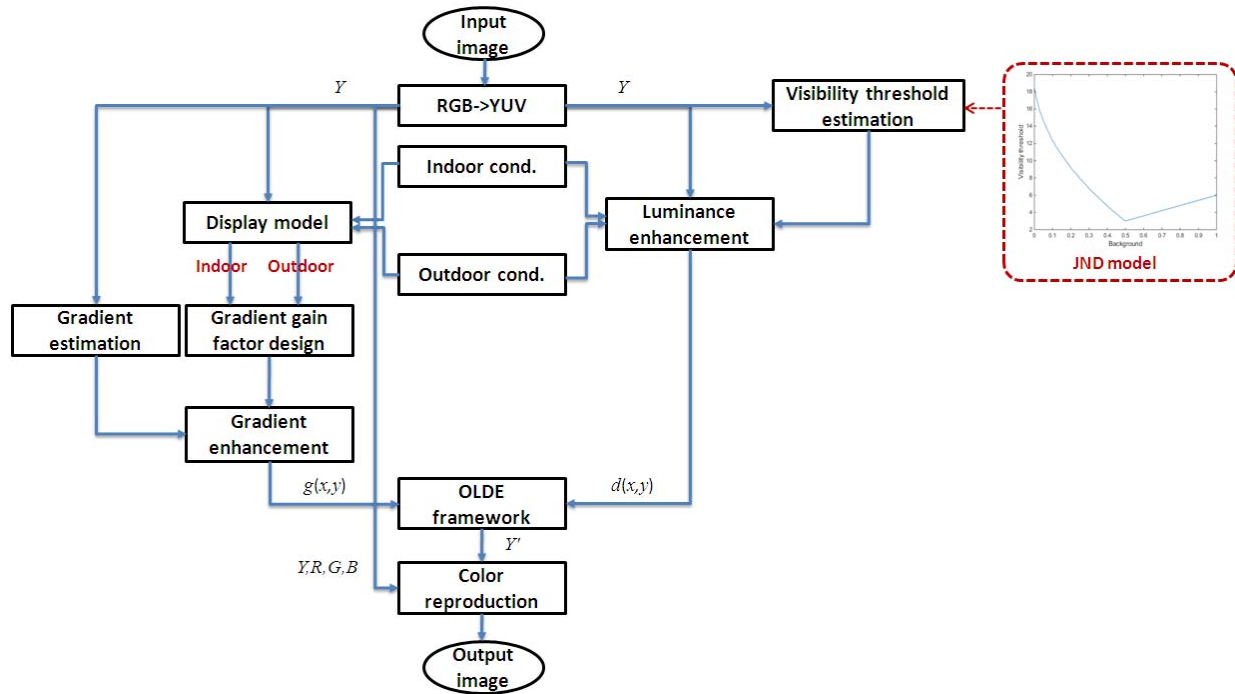


Fig. 1. Proposed weighted optimization framework for ambient light adaptive display enhancement. Visibility threshold is calculated by background luminance masking model in just noticeable difference (JND) model. OLED framework means optimized luminance and detail enhancement framework in Eq. (3). In general, the magnitude of ambient light in the indoor environment is smaller than 1000lux, e.g. library and office: 300-500lux, and thus we regard the environment with the ambient light larger than 1000 lux as the outdoor environment. The magnitude of ambient light is measured by a lux sensor.

Local contrast was defined as the ratio of pixel luminance to its local surround. However, [11] leads to noise artifacts emerge in textural regions.

Moreover, illumination-level adaptive color reproduction for mobile displays [12] introduced an S-shaped Naka-Rushton equation [13] for luminance enhancement and chroma compensation for color distortions. This method achieved bright and colorful images but introduced a color cast problem. Mantiuk *et al.* [14] combined an HVS model for ambient light with HDR tone mapping to realize ambient light adaptive tone mapping by minimizing visual distortions. Despite displayed images were adjusted based on display properties and ambient light, image appeared blurred and lost details. Aydın *et al.* [15] applied maladaptation and temporal recovery of the sensitivity mechanism to the previous HVS model. The maladaptation is caused by dramatic changes in lighting condition as well as seeing very bright and dark objects. This model has been used for visibility and detail loss prediction in the car interior displays under varying lighting conditions. Kim *et al.* [16] developed the adjusted contrast sensitivity function (CSF) for surroundings, and then used it as a filter to enhance small sized mobile LCDs according to surround luminance. However, [16] produces distortions of contrast and details in bright regions. Yang proposed a multiscale decomposition-based method to compensate perceived image contrast in different local bands by using the same CSF [17]. However, [17] does not provide good enhancement of luminance and contrast under strong ambient light. Xue and Chen [18] developed a mobile video perception model in different viewing conditions and used it as an assessment tool for perceptual video coding. The viewing conditions considered display size, viewing distance, ambient

light, and body movements. Wanat and Mantiuk proposed luminance retargeting which improves the image appearance under a given brightness range in screen displays [19]. They considered the effect of adapting luminance on the contrast sensitivity of the human visual system (HVS), and employed a steady-state CSF function in [28]. However, the steady-state CSF only considers adapting luminance (i.e., displayed luminance) and spatial frequency as input. Ambient light has an effect only on the display reflection which is added to the displayed luminance. Therefore, this method is not effective in considering the effect of ambient light on the adaptation state of HVS.

B. Contributions

Ambient light often causes not only contrast distortions but also brightness reduction in HVS. Especially, in the strong ambient light, brightness reduction is more serious, which causes dark perception as shown in Fig. 3. However, only luminance enhancement leads to detail loss by a non-linear transfer function (e.g. gamma function) which enhances contrast and increases luminance in a low intensity range while compressing the dynamic range in a high intensity range. Thus, ambient light affects details in displayed images. In this paper, we propose adaptive enhancement of both luminance and details in images under ambient light. We combine the data term for luminance enhancement and the gradient term for detail enhancement into a weighted optimization framework. First, we perform luminance enhancement using an ambient light nonlinear intensity-transfer function. This function increases luminance in an image, and thus enhances the human visual response reduced by ambient light.

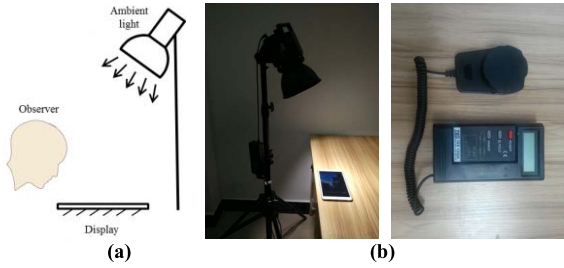


Fig. 2. (a) Problem definition of the proposed method and experimental setup. (b) Experimental equipment (photographic lamp, tablet, and lux sensor).

Second, we compensate for gradient distortions caused by ambient light based on TVIA function [26]. Finally, we achieve readability enhancement of displayed images by solving the optimization framework based on a numerical solver. Experimental results demonstrate that the proposed method effectively enhances luminance in displayed images while successfully preserving details under ambient light. The framework of the proposed method is illustrated in Fig. 1. Compared to existing methods, main contributions of this work are as follows:

- (1) We provide both luminance and detail enhancement in displayed images under ambient light based on a weighted optimization framework;
- (2) We design an ambient light adaptive nonlinear transfer function for luminance enhancement considering the HVS model, display properties, image contents, and ambient light;
- (3) We employ a threshold versus intensity adaptation (TVIA) function under ambient light for detail enhancement, and adaptively control the weight in the optimization framework according to gradient distortion of luminance enhancement;
- (4) We improve the readability of mobile displays without elevating backlight, which achieve power consumption.

The remainder of this paper is organized as follows. In Section II, we explain the proposed method in detail. Experimental results and their corresponding analysis are provided in Section III, and we draw conclusions in Section IV.

II. PROPOSED METHOD

In general, HVS perceives displayed images adapting to ambient light as shown in Fig. 2(a). Thus, we assume that ambient light has effects on both display reflection and human visual perception. Ambient light strength in outdoor environment is often much larger than the displayed luminance, which seriously affects human visual perception, i.e. darkly perceived [13]. Thus, we consider both factors of ambient light and displayed luminance in the perceptual model. Strong ambient light causes dark perception of displayed images in HVS. Its main reason is the lightness adaptation mechanism in HVS. Lightness adaptation means HVS shifts the middle range ($10^1 \text{cd/m}^2 \sim 10^4 \text{cd/m}^2$ in Fig. 3) in response curve of their photoreceptors rightwards or leftwards to adapt to ambient light, which makes human eyes more sensitive to the luminance around the middle range.

Based on this mechanism, human eyes adapt to the viewing condition and effectively perceive natural scenes. However,

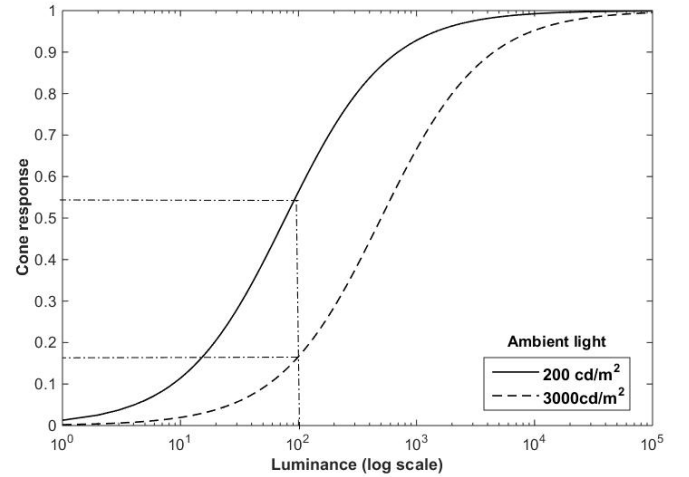


Fig. 3. The cone response of displayed images on mobile displays under 200cd/m^2 (solid line) and 3000cd/m^2 (dash line). We assume that the maximum luminance of the display is 100cd/m^2 . It can be observed that maximum response decreases from 0.55 to 0.15 when mobile display moves from indoor (200cd/m^2) to outdoor environment (3000cd/m^2). The relationship between ambient light (lux) and the adaptation level (cd/m^2) is shown in Eq. (2).

when mobile displays are used in the outdoor environment, human eyes adapt to strong ambient light but not sensitive to the low intensity emitted from mobile displays. Thus, the displayed images are darkly perceived by human eyes due to the limited dynamic range of displays as illustrated in Fig. 3. The cone response of HVS is calculated by the S-shaped Naka-Rushton equation [13] as follows:

$$R_{\text{cone}} = \frac{Y^n}{Y^n + \delta^n} \quad \delta = I_A^\alpha \times \beta \quad (1)$$

where Y is displayed image luminance; δ is half-saturation constant; n is the sensitivity constant with the range from 0.7 to 2.0; α and β are constants and different for rods and cones (α is 0.69 and β varies from 5.83 to 2.00) [20]; and I_A is the adaptation level. In Eq. (1), I_A is calculated based on the CIECAM 97s to consider the adaptation to ambient light reflected from surrounding objects as follows [21]:

$$I_A = \frac{R \cdot E_{\text{amb}}}{\pi} \quad (2)$$

where E_{amb} is ambient light intensity (lux) obtained by a lux sensor; R is the reflectance; and π is the circular constant. In this work, we assume there exists a perfectly reflecting diffuser, and thus set R to 1.0 [21]. It can be observed that the maximum response greatly decreases when mobile displays moves from indoor (200cd/m^2) to outdoor (3000cd/m^2) environment. This is the reason why displayed images in the outdoor are perceived more darkly. Also, the response in the dark region ($0\text{--}10 \text{cd/m}^2$) of images has been degraded largely, which decreases approximately to 0. Thus, perceived luminance distortion becomes very severe under strong ambient light and image luminance should be enhanced to elevate the human visual response for improving image visual quality, especially in the dark regions. Thus, it is required to enhance the luminance of images. However, image details are also lost after luminance enhancement. Thus, it is required to enhance

both luminance and details of displayed images under ambient light.

Bhat *et al.* [22] have proposed the gradient-domain optimization framework which adopts gradient constraints to improve visual appearance of results in data term as follows:

$$E(f(x, y)) = \int \int (f(x, y) - d(x, y))^2 + \lambda \cdot A(x, y) \cdot \|\nabla f(x, y) - \mathbf{g}(x, y)\|^2 dx dy \quad (3)$$

where $d(x, y)$ represents the data term for luminance enhancement; and $g(x, y)$ is the gradient term under ambient light; λ balances between two terms; $A(x, y)$ means spatially varying weights which adjust detail enhancement; and $f(x, y)$ is the enhanced result generated by minimizing the energy function of Eq. (3). This framework has been successfully used in some applications such as visualization for high dynamic range infrared images [23], saliency sharpening, and de-blocking. In the proposed optimization framework, it is required to enhance both luminance and details of displayed images under ambient light. We obtain $d(x, y)$ by an ambient light nonlinear intensity-transfer function based on the HVS model, display properties, image contents, and ambient light. For $g(x, y)$, we introduce TVIA functions to design the gradient gain factor under different ambient lights, which compensates for gradient distortions caused by ambient light. We use the enhanced image luminance as $A(x, y)$ in the optimization framework to minimize gradient distortions caused by luminance enhancement.

A. Luminance Enhancement

We provide an ambient light adaptive nonlinear transfer function to enhance luminance of displayed images under ambient light. We perform luminance enhancement in the ambient light adaptive nonlinear transfer function depending on display properties, ambient light, and HVS model. We calculate the visibility threshold for each pixel using the background luminance masking model derived from Just noticeable difference (JND) [24]. Given the input image, we achieve adaptive luminance enhancement under ambient light as follows:

$$d(x, y) = (L(x, y))^{i+(1-i) \cdot s(x, y)} \quad (4)$$

where $L(x, y)$ is luminance of the original image; and $d(x, y)$ is its enhanced image by an adaptive nonlinear transfer function; and i controls maximum enhancement degree depending on display properties, reflected light generated from ambient light and HVS response in the indoor as follows:

$$i = \frac{\log(L_{\max} + L_{\text{refl}}) - \log(L_{\text{black}} + L_{\text{refl}})}{\log(f_{\max}(R_{\text{indoor}})) - \log(f_{\min}(R_{\text{indoor}}))} \quad (5)$$

where R_{indoor} is the human visual response in the indoor by Eq. (1); L_{refl} represents the physical reflected luminance (units: cd/m^2) by Eq. (11); $f_{\max}$ and $f_{\min}$ determine maximum and minimum values, respectively. If we only use a simple power function $d = L^i$ for luminance enhancement, image contrast would be largely reduced and image luminance is over enhanced as shown in Fig. 4(b). Thus, we introduce an adaptive parameter $s(x, y)$ to control the adaptive enhancement

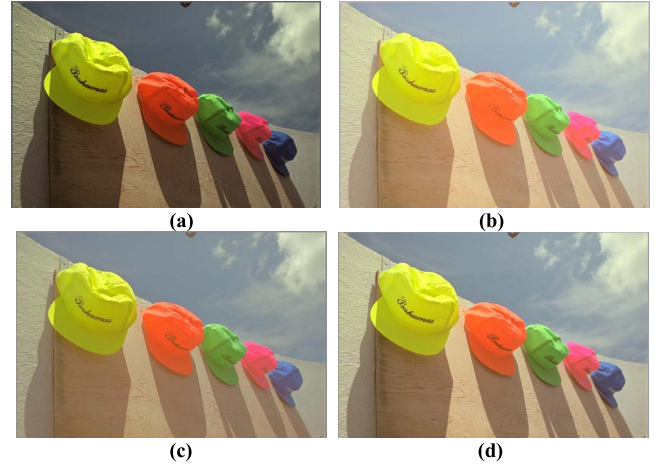


Fig. 4. Luminance enhancement by different nonlinear transfer functions. (a) Original image. (b) Simple power function: $d = L^i$. (c) Adaptive power function guided by image luminance: $d = L^{i+(1-i)*L}$. (d) Proposed adaptive power function $d = L^{i+(1-i)*L*T}$. (Ambient light: 10000lux).

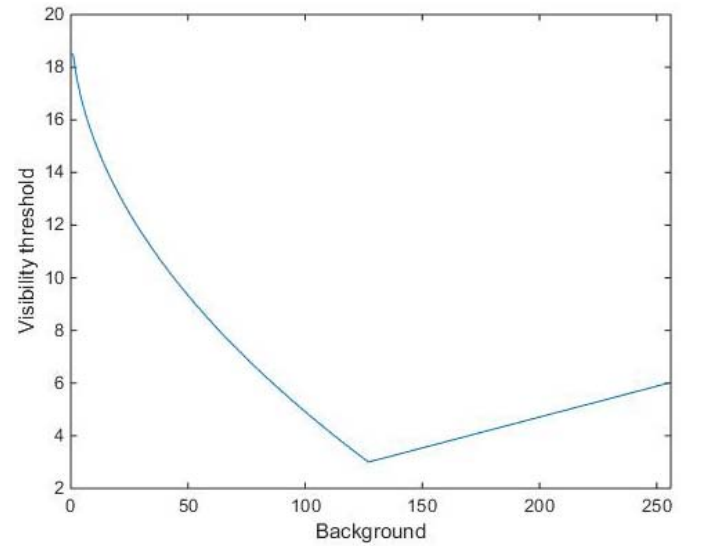


Fig. 5. Visibility threshold according to the background luminance

degree based on image luminance L and background luminance masking T as follows:

$$s(x, y) = L(x, y) \cdot T(x, y) \quad (6)$$

where $L(x, y)$ is the normalized gray level of each pixel (i.e. 0-1) in the original image; and $T(x, y)$ represents the visibility threshold to avoid the over-enhanced problem. By substituting Eq. (6) into Eq. (4), we provide a new adaptive nonlinear transfer function as follows:

$$d(x, y) = (L(x, y))^{i+(1-i) \cdot T(x, y) \cdot L(x, y)} \quad (7)$$

In general, ambient light degrades images in dark regions more seriously than bright ones as shown in Fig. 3. Thus, we first introduce $L(x, y)$ as a parameter to control the adaptive enhancement degree, which leads to enhance in dark regions more than in bright ones. However, large enhancement in dark regions causes reduction of a dynamic range in the middle intensity range of [0.2, 0.5], which makes the enhanced image tone look unnatural as shown in Fig. 4(c). Thus, we add

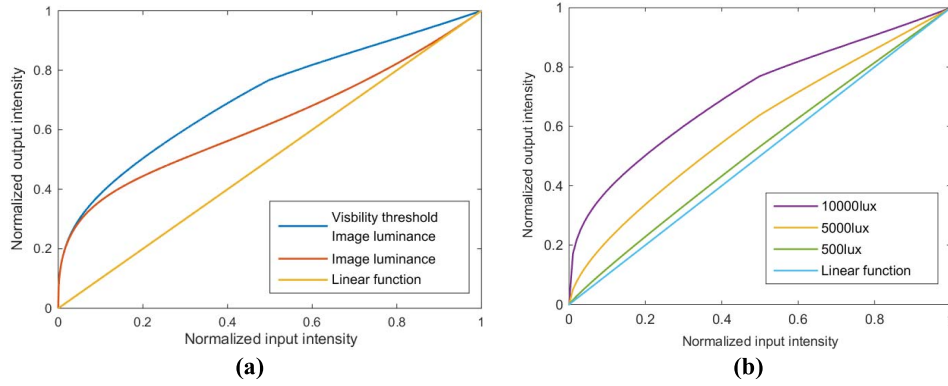


Fig. 6. Intensity mapping curves. (a) Different nonlinear transfer functions which considers luminance (red curve), visibility threshold and image luminance (blue curve) compared with linear function (yellow curve) when ambient light is 10000lux. (b) Proposed nonlinear transfer functions under ambient light.

visibility threshold $T(x, y)$ to the nonlinear transfer function to avoid the over-enhancement problem in Eq. (7), and thus the enhanced image looks natural and bright as shown in Fig. 4(d). We calculate $T(x, y)$ using the background luminance masking model by Eqs. (8) and (9). Background luminance masking is the phenomenon that HVS is sensitive to luminance contrast rather than absolute luminance [24],[25].

It is required that we enhance luminance less in the region of the high visibility threshold than that of the low visibility threshold.

$$T(x, y) = \begin{cases} 17 \left(1 - \sqrt{\overline{L(x, y)}/127} \right) + 3, & \text{if } \overline{L(x, y)} \leq 127 \\ \frac{3}{128} (\overline{L(x, y)} - 127) + 3, & \text{otherwise} \end{cases} \quad (8)$$

$$\overline{L(x, y)} = \frac{1}{32} \sum_{i=1}^5 \sum_{j=1}^5 L(x-3+i, y-3+j) \cdot B(i, j) \quad (9)$$

where $\overline{L(x, y)}$ is an average background luminance; and $B(i, j)$ is a weighted low-pass filter [24]. Equation (8) measures the relationship between visibility threshold and the background luminance by subjective experiments [25]. The visibility threshold for each grey level is measured when the superimposed stimuli contaminated in flat regions are just noticeable. The visibility threshold is shown by $T(x, y)$ in Eq. (8). Based on the subject experiments, it has been discovered that HVS is sensitive to 128, i.e. the lowest visibility threshold, and follows the curve in Fig. 5 called luminance adaptation. Given a background luminance $\overline{L(x, y)}$, HVS cannot detect the foreground luminance when its difference from $\overline{L(x, y)}$ is within $T(x, y)$. Equation (9) simulates the flat regions by calculating the average of the pixels within the neighboring portion of the image [25]. This model indicates the variation of visibility threshold with background luminance in Fig. 5. High visibility threshold means low sensitivity, which means a large number of noise level and contrast cannot be perceived. The enhancement in these regions should be small to avoid making much noise and undesirable contrast visible. We apply the normalized $T(x, y)$ to the proposed nonlinear transfer function.

TABLE I
NOTATIONS OF MAIN VARIABLES IN SECTION II.B

Notation	Definition	Unit/Range
$L(x, y)$	The gray level of each pixel from RGB→YUV.	0-1
L_a	The adaptation level to ambient light (surrounding luminance)	cd/m ²
$L_d(x, y)$	Displayed luminance	cd/m ²
$L_o(x, y)$	The retinal luminance	cd/m ²
$\Delta L(L, L_a)$	The detection threshold of luminance difference considering L and L_a	cd/m ²
$T_e(L, L_a)$	The threshold elevation for detection contrast ($\Delta L/L$) due to maladaptation	--
$s(x, y)$	Spatially varying gain factors for image gradient under ambient light	--

We obtain intensity mapping curves by different nonlinear transfer functions with various factors under ambient light as shown in Fig. 6(a). The proposed method in the blue curve of Fig. 6(a) enhances luminance in dark regions while preserving contrast in the middle intensity range of [0.2, 0.5] compared with the red curve which only considers luminance for enhancement. Thus, we produce different intensity mapping curves under different ambient lights as shown in Fig. 6(b).

B. Ambient Light Image Gradient Enhancement

After luminance enhancement, the enhanced image suffers from significant detail loss under ambient light. The notations of main variables in Section II.B are described in TABLE I. First, we analyze gradient distortion under ambient light from three viewpoints: reflected light, HVS perception, and luminance enhancement. Based on the comprehensive analysis, we employ TVIA function to compensate for the perceived gradient distortion under ambient light, and provide a weighted optimization framework considering the gradient distortion caused by luminance enhancement.

1) *Reflected Light From Ambient Light*: We introduce a display model to consider how ambient light affects displayed images, which effectively considers display properties, and check whether the reflected light affects image gradient. We produce displayed images under ambient light by the following display model [14]:

$$L_d(L) = L^\gamma \cdot (L_{max} - L_{black}) + L_{black} + L_{refl} \quad (10)$$

where L_{max} and L_{black} are peak luminance and black level of the output display (L_{black} : the luminance of the black pixel displayed in a perfectly dark room), respectively; γ is display gamma (usually 2.2); and L_{refl} is the reflected light emitted from display surface as follows [14]:

$$L_{refl} = \frac{k}{\pi} E_{amb} \quad (11)$$

where E_{amb} is ambient light (lux) measured by a lux sensor and k is display reflectivity. In this work, specular reflection is not considered due to the assumption of non-glossy screens, which has the same definition as a display model in [14].

The display model in Eq. (10) indicates that ambient light increases luminance in images by adding same reflected light intensity. Thus, the reflected light does not affect the original image gradient calculated by difference between neighboring pixels.

2) *Effect of Ambient Light on HVS*: We analyze perceived gradient distortion (i.e. detail loss) under ambient light based on the phenomenon that luminance of displayed images increases as ambient light increases (discussed in Section II.B.1). To estimate gradient distortions, near-threshold models such as Weber's law, CSF, and JND, have been commonly used.

Although Weber's law effectively represents the minimum noticeable difference in HVS when an observer adapts to the background luminance [13], HVS is not able to optimally adapt to the displayed luminance (background) under ambient light. This is because ambient light is not a neglected factor which is much larger than displayed luminance and seriously affects human perception of displayed luminance. Therefore, we consider two factors of displayed luminance and ambient light in our perception model. We adopt the maladaptation derivative of the threshold versus intensity (TVI) model [26] to consider gradient distortions under ambient light, i.e. threshold versus intensity adaptation (TVIA) function. We utilize TVIA function as the luminance difference threshold $\Delta L(L, L_a)$ to consider both displayed and adapting luminance. TVIA function is a generalized form of TVI, which is the same as TVI curve when $L_a = L$. We obtain TVIA function from human visual response as follows:

$$\Delta L(L_o, L_a) = tvi(L_o) T_e(L_o, L_a) \quad (12)$$

$$T_e(L, L_a) = \frac{\Delta L(L, L_a)/L}{\Delta L(L_a, L_a)/L_a} = \frac{(L^n + L_a^n)}{4L^n L_a^n} \quad (13)$$

where L_a is the degree of adaptation to ambient light in Eq. (2), which affects human perception seriously under ambient light; L_o is the retinal luminance caused by glare effects, which is calculated by the convolution of an optical transfer function [27] and the displayed image luminance L_d ; and n is the sensitivity constant in Eq. (1). The retinal luminance L_o , caused by glare effects, is calculated by the convolution of an optical transfer function OTF(ρ, d) and the displayed image luminance L_d . The optical transfer function in [27] is obtained based on a pupil diameter d in mm and spatial frequency ρ in cycles per degree as follows:

$$OTF(\rho, d) = \exp \left[- \left(\frac{\rho}{20.9 - 2.1d} \right)^{1.3 - 0.07d} \right] \quad (14)$$

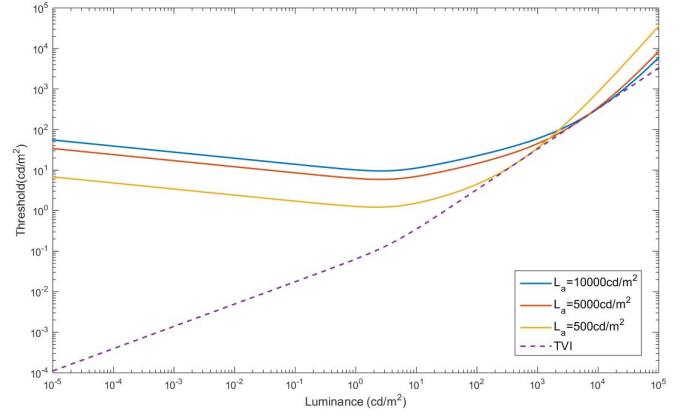


Fig. 7. TVIA curves according to surrounding luminance L_a . TVI is generally located below TVIA curves.

The pupil diameter d depends on surrounding luminance L_a , which is the adaptation level of ambient light in Eq. (2) as follows:

$$d = 4.9 - 3 \tanh [0.4 (\log L_a + 1.0)] \quad (15)$$

We obtain TVIA function based on the CSF in [28]. Fig. 7 shows TVIA curves under different surrounding luminance (i.e. L_a in Eq. (13); yellow: 500 cd/m²; red: 5000 cd/m² and blue: 10000 cd/m²). As shown in the figure, TVI is generally located below TVIA curves. TVIA curves have two main characteristics: 1) When the luminance is lower than 10 cd/m² (much smaller than three surrounding luminance), the threshold is changed with the same slope according to the luminance. Also, the threshold in low surrounding luminance is smaller than that in high surrounding luminance. This characteristic shows large gradient distortion in low luminance (<10cd/m²) because of the same space between two curves under different ambient lights. 2) When the luminance is beyond 10 cd/m², the threshold of luminance increases rapidly and TVIA curve intersects with the other curves. The closer surrounding luminance is, the more rapidly the threshold increases. This characteristic indicates that the gap between two curves becomes narrow and the small gradient distortion exists. Therefore, the gradient distortion in dark regions is larger than that in bright ones when displayed images are moved from indoor to outdoor environment. We obtain the gradient in the enhanced image as follows:

$$\mathbf{g}(x, y) = \nabla \mathbf{L}(x, y) s(x, y) \quad (16)$$

$$s(x, y) = \frac{\Delta L(L_{outdoor}, L_{a-outdoor})}{\Delta L(L_{indoor}, L_{a-indoor})} \quad (17)$$

where $\mathbf{g}(x, y)$ is the gradient of the enhanced image under ambient light; $\nabla \mathbf{L}$ is the gradient in the original image; and $s(x, y)$ represents the gain factor for $\nabla \mathbf{L}$, which depends on ambient light and TVIA model; L_{indoor} and $L_{outdoor}$ are retinal luminance of displayed images in the indoor and outdoor environments, respectively; $L_{a-indoor}$ and $L_{a-outdoor}$ are surrounding luminance in the indoor and outdoor environments, respectively, which are calculated by Eq. (2). Fig. 8(b) shows $s(x, y)$. Large $s(x, y)$ indicates that the detection threshold increases largely compared with the previous environment, which represents a large gradient distortion by HVS. It is



Fig. 8. Gradient gain maps in *Hat* under ambient light (10000lux). (a) Original image in luminance (gray level). (b) Its gradient gain map $s(x, y)$.

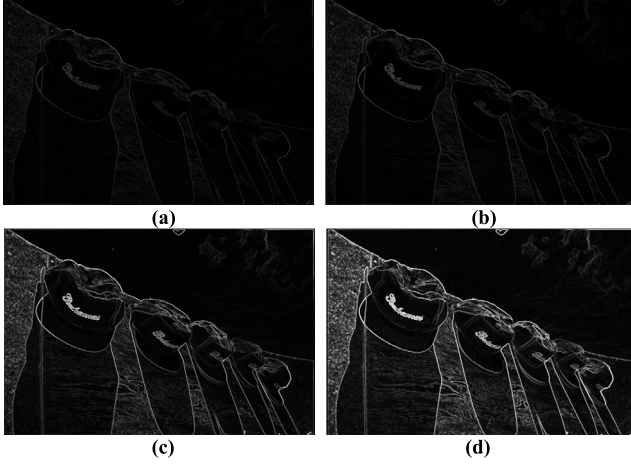


Fig. 9. Enhanced gradient maps in *Hat* under different ambient lights. (a) Original gradient. (b) 500lux. (c) 50000lux. (d) 10000lux.

required to enhance the region with the large gradient distortion. As shown in Fig. 8(b), the gradient distortion in dark regions is larger than that in bright ones. The enhanced gradient under different ambient lights is shown in Fig. 9. As shown in the figure, the proposed method achieves a good performance in gradient enhancement under strong ambient light.

3) *Effect of Luminance Enhancement on Gradient Distortions*: The proposed method enhances luminance, but reduces gradient in displayed images. As shown in Fig. 6(b), the luminance enhancement function mainly focuses on the enhancement of dark regions where the slope of the function is larger than 1 in them.

However, the slope of the function gradually becomes less than 1 with the increase of luminance. This phenomenon leads to more gradient reduction in the bright region than in dark one. To compensate for the distortions, we choose the weights of the optimized framework to guide detail enhancement. The large weight is needed for large detail enhancement in the bright region while the small weight is needed in the dark region. Thus, we use the enhanced luminance as the weight $A(x, y)$ for the optimized framework to guide detail enhancement as follows:

$$A(x, y) = d(x, y)^\alpha \quad (18)$$

where $d(x, y)$ is the luminance in the enhanced image; and α controls the sensitivity to image luminance ($1 < \alpha < 2$). Fig. 10 shows the enhanced gradient in *Guy* by different α in Eq. (18). It can be observed that when $A(x, y) = 1$,

the enhanced gradient maps in Figs. 10(b) and 11(b) cause noise amplification in dark regions and detail reduction in bright regions compared with the original image. α adjusts the decrease of $A(x, y)$ in low intensity, which reduces detail enhancement to alleviate the noise amplification as shown in the left pictures of Figs. 11(c)-(d). With the increase of α , $A(x, y)$ decreases, and thus the result in Fig. 11(d) has smaller noise than that in Fig. 11(c) has. In high intensity, $A(x, y)$ slightly decreases as α increases, and thus the gradient in bright regions is well-preserved as shown in the right pictures of Figs. 11(c)-(d). Therefore, the gradient in bright regions is preserved, while the gradient in dark regions is attenuated to avoid noise amplification with the increase of α .

C. Numerical Solution

Given $d(x, y)$ and $g(x, y)$ under ambient light, f is obtained by minimizing the weighted optimization framework in Eq. (3), which achieves the optimum results by balancing between $d(x, y)$ and $g(x, y)$. Most optimization methods convert the energy function into the sparse linear system [29], and then apply a conjugate-gradient method to solve this iteratively. We use the solver in [30] to minimize the energy function. We rewrite Eq. (3) using the matrix notation as follows:

$$(f - d)^T (f - d) + \lambda \left[(D_x d - \mathbf{g}_x)^T A (D_x d - \mathbf{g}_x) + (D_y d - \mathbf{g}_y)^T A (D_y d - \mathbf{g}_y) \right] \quad (19)$$

where d is the enhanced image luminance and f is its resulted image by the optimized framework in Eq. (19); A is diagonal matrices which contain the weights of the optimized framework; $\mathbf{g}_x$ and $\mathbf{g}_y$ correspond to partial derivatives with respect to enhanced image gradient $\mathbf{g}$; D_x and D_y are discrete differentiation operators. We obtain f by minimizing (3), which is the solution of the linear system as follows:

$$\begin{aligned} & \left(I + \lambda (D_x^T A D_x + D_y^T A D_y) \right) f \\ &= d + \lambda (D_x^T A \mathbf{g}_x + D_y^T A \mathbf{g}_y) \end{aligned} \quad (20)$$

$$\begin{aligned} f &= \left(I + \lambda (D_x^T A D_x + D_y^T A D_y) \right)^{-1} \\ &\quad \times \left(d + \lambda (D_x^T A \mathbf{g}_x + D_y^T A \mathbf{g}_y) \right) \end{aligned} \quad (21)$$

In this work, D_x and D_y are forward difference operators, while D_x^T and D_y^T are backward ones. We obtain the final f by Eq. (21).

D. Color Reproduction

We reproduce the enhanced color image M_e from f as follows [31]:

$$M_e(x, y) = M_o(x, y) \left(\frac{L_e(x, y)}{L_o(x, y)} \right)^\tau \quad (22)$$

where $M_o(x, y)$ are trichromatic channels ($o \in \{r, g, b\}$) of the original image; $L_e(x, y)$ is the enhanced image which is resulted image f in Eq. (19); $L_o(x, y)$ is original luminance;



Fig. 10. Enhanced gradient maps in *Guy* according to different α in Eq. (3). (a) Original image. (b) $\alpha = 0$. (c) $\alpha = 1.0$. (d) $\alpha = 2.0$. Ambient light is 10000lux and λ is 0.2.

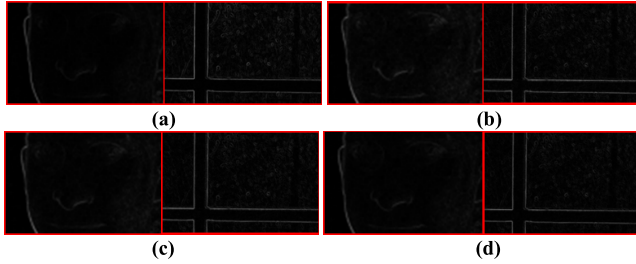


Fig. 11. The magnified area for enhanced gradient maps in Fig. 10. (a) Original image. (b) $\alpha = 0$. (c) $\alpha = 1.0$. (d) $\alpha = 2.0$.

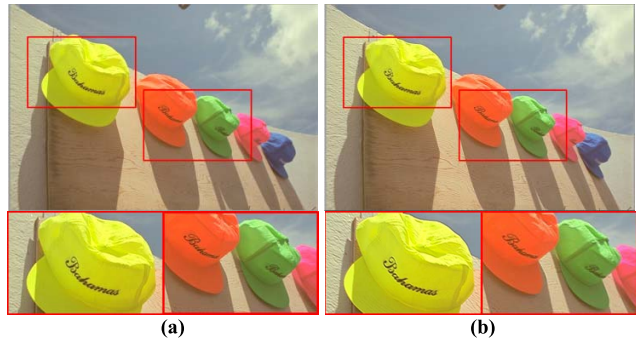


Fig. 12. Enhanced color image in *Hat* under ambient light (10000lux). (a) Luminance enhancement in Eq. (7). (b) Luminance and detail enhancement in Eq. (19).

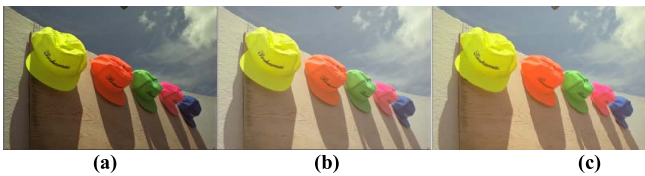


Fig. 13. Enhanced color image in *Hat* under ambient light (10000lux) with different τ . (a) Original image. (b) Enhanced image with $\tau = 0.6$. (c) Enhanced image with $\tau = 0.8$.

τ varies between 0.6 and 0.8, and we adjust it to obtain the best visual quality of the enhanced color image. Fig. 11 shows the enhanced color image by only luminance enhancement in Eq. (7) and the proposed method which achieves luminance and detail enhancement in Eq. (19). It can be observed that the details in Fig. 12(b) have been enhanced and successfully appeared compared to Fig. 12(a) processed by only luminance enhancement in Eq. (7). The color reproduction reproduces the color of the original image while producing saturated colors in the enhanced results with variation of τ as shown in Fig. 13, which compensates for the color distortion under ambient light.

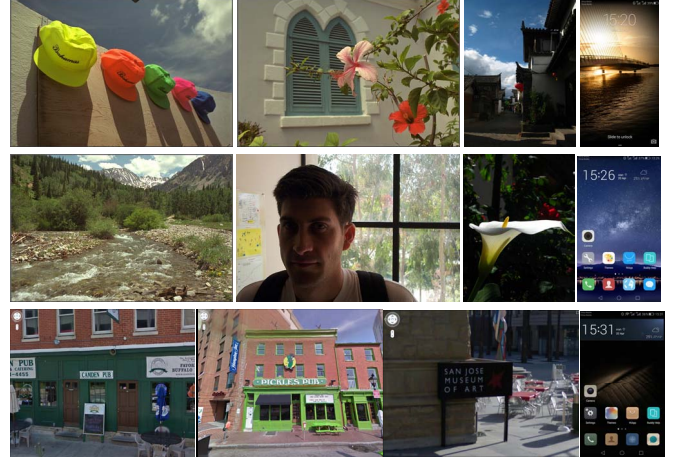


Fig. 14. Test images for experiments: *Hat*, *Window*, *Alley*, *Mobile Screen 1*, *Forest*, *Guy*, *Flower*, *Mobile Screen 2*, *Pub*, *House*, *Billboard*, *Mobile Screen 3*.

III. EXPERIMENTAL RESULTS

To evaluate the performance of the proposed method, we perform experiments on 12 test images. Six images are from the Kodak lossless true color image suite [32] and images captured by a single lens reflex camera (Canon EOS 60D): *Hat*, *Window*, *Alley*, *Forest*, *Guy*, and *Flower*. The resolution of *Hat*, *Window* and *Forest* is 768×512 , that of *Alley* and *Flower* is 1824×1368 , and that of *Guy* is 592×896 . The six text images are divided into two groups: three text content images [33] with resolution of 1024×768 and three screen content images from mobile phone with resolution of 1080×1920 . Fig. 14 shows the test images for experiments. For the tests, we use a PC with Intel (R) Core (TM) i5 CPU (2.60GHZ) and 4.00GB RAM running a Windows 7 environment and MATLAB. We use Samsung GALAXY Tab S T700 as the display for experiments, which has maximum luminance $L_{\max} = 544 \text{cd/m}^2$ and the reflectivity of 4.7% [34]. We utilize a photographic lamp (DOF LED-1000) to generate ambient light, where testing lights are 500lux, 5000lux, and 10000lux which represent environments of office, shadows in the cloudy day, and cloudy daylight, respectively.

To measure ambient light, we use a lux sensor of TES 1332A as shown in Fig. 2(b). Fig. 2 illustrates our experimental setup. As shown in Fig. 2, the photographic lamp is used to project lights onto the whole display surface and viewers, which simulates displayed images under ambient light. The viewer's position is illustrated in Fig. 2(a). The distance from the photographic lamp to displays simulates the ambient light environment. The glare luminance in viewer's eyes is

TABLE II
SUBJECTIVE EVALUATIONS OF THE PROPOSED METHOD, [12], [14], [16]

500lux					5000lux					10000lux				
[12]	[14]	[16]	Proposed	Original	[12]	[14]	[16]	Proposed	Original	[12]	[14]	[16]	Proposed	Original
1.3	4.3	0.5	6.0	0	1.0	3.8	0.5	6.8	0	0.5	1.5	1.3	8.8	0



Fig. 15. Experimental results in *Hat*. (a) Original images. (b)-(d) Enhanced color images under 500lux, 5000lux, and 10000lux. From the first row to last row: [12], [16], [14], and the proposed method.

generated by the reflected lights from ambient light. In our experiments, the angle between viewers and specular reflected light is quite large, and thus specular reflection doesn't occur.

A. Visual Comparisons

Ten observers, who consist of experts in image processing, participate in evaluating the visual quality of the enhanced results by different methods. The average age of them is 24 years old; ages range from 22 to 25 years old; and two observers are female. We test the proposed method and other two ones under different ambient lights: 500lux, 5000lux, and 10000lux. We set λ in Eq. (19) from 0.1 to 0.3. We set the maximum luminance of the tablet to 100cd/m², and don't use the auto-brightness function in the experiments. Each observer selects an image which has the best readability among four results by four methods (the proposed method, [12], [14], [16]) and the original image under one

ambient light. Here, we define the readability as two parts: 1) the best luminance and contrast of image under ambient light; 2) the least distortion. For experiments, each picture is randomly presented to observers in a light condition, and the total number of experiments are 36 (12 test images and 3 light conditions).

We compute the average score, i.e. the number of selected times for each method, as listed in TABLE II. The larger average score is, the better visual quality the method achieves under ambient light. The proposed method outperforms [12] and [16] in performance which have serious luminance distortions under high ambient light. [14] achieves the strongest contrast enhancement among four methods, but causes much distortion in the dark and bright regions. Moreover, the distortions gradually become more serious as ambient light increases. Experimental results show that the proposed method significantly improves the readability



Fig. 16. DRI evaluation results of four methods when ambient light is 10000lux. Left to right: [12], [16], [14], and the proposed method. Green pixels represent loss of visible contrast, while red ones represent reversal of visible contrast.

of displayed images with the least distortion under varying ambient light environment, especially strong ambient light. Fig. 15 shows experimental results of the proposed method compared with [12], [14], and [16]. In [12], luminance compensation by an HVS model and chroma compensation by a color appearance model were performed to make displayed images more vivid. The chroma compensation enhances the luminance compensation result, and makes the results look bright and colorful due to the saturation effect in the color appearance model. From the first row in Fig. 15, it can be observed that the enhanced result looks colorful and bright with the increase of ambient light by [12]. However, there is a color cast problem in the enhanced image. This is because only luminance adjustment changes the original ratio of three channels and makes images reddish or greenish as shown in the first row of Figs. 15(c) and (d). [16] compensates for the distortions of contrast sensitivity between different ambient lights by a surrounding CSF. It preserves image contrast in spatial frequencies when we use mobile devices from indoor to outdoor. However, it causes distortions of contrast and details in bright regions, and thus over-enhances the luminance as shown in the second row of Fig. 15. Reference [14] shows better luminance and contrast in the results, but often causes distortions in some regions because of limited dynamic range of display as shown in the third row of Fig. 15. The proposed method achieves enhancement in both luminance and details. In practice, the luminance distortion gradually becomes more serious than the other distortions as ambient light increases. The proposed method uses an ambient light non-linear transfer function for luminance enhancement, which improves the image readability and avoids too much image distortion. If luminance enhancement is enough, especially in

the strong ambient light, lost image contrast and details are appeared. Thus, the proposed method obtains better visual quality than [12] and [16]. We also enhance image details under ambient light by minimizing detail loss and gradient distortions. Thus, we achieve less distortion than [14]. Figs. 17-19 show experimental results in *Billboard*, *Alley*, *Mobile Screen 1*, and *Guy* by four methods, respectively. As shown in the figure, with the increase of ambient light, the proposed method remarkably enhances both luminance and details in images, and improves their readability with the least distortion under varying ambient light.

B. Objective Evaluations

The proposed method aims at enhancing the readability of displayed images with the least distortion. However, evaluation metrics such as peak signal to noise ratio (PSNR), structural similarity index (SSIM) [39], and visual information fidelity (VIF) [40] are not suitable for assessing displayed images under ambient light because they are not considering human visual perception under ambient light in evaluations. Therefore, we perform three kinds of objective evaluations considering human visual perception as follows:

1) *Evaluation of luminance and contrast enhancement*: We use absolute mean brightness error (AMBE) [35], discrete entropy (DE) [36], and measure of enhancement (EME) [37] to evaluate luminance and contrast enhancement. AMBE measures the mean increment of luminance between the enhanced and original images, DE represents the amount of information in an image, and EME evaluates an average contrast of an image. Higher AMBE, DE, and EME indicate better readability of displayed images under ambient light in terms of luminance and contrast enhancement. TABLE II lists average

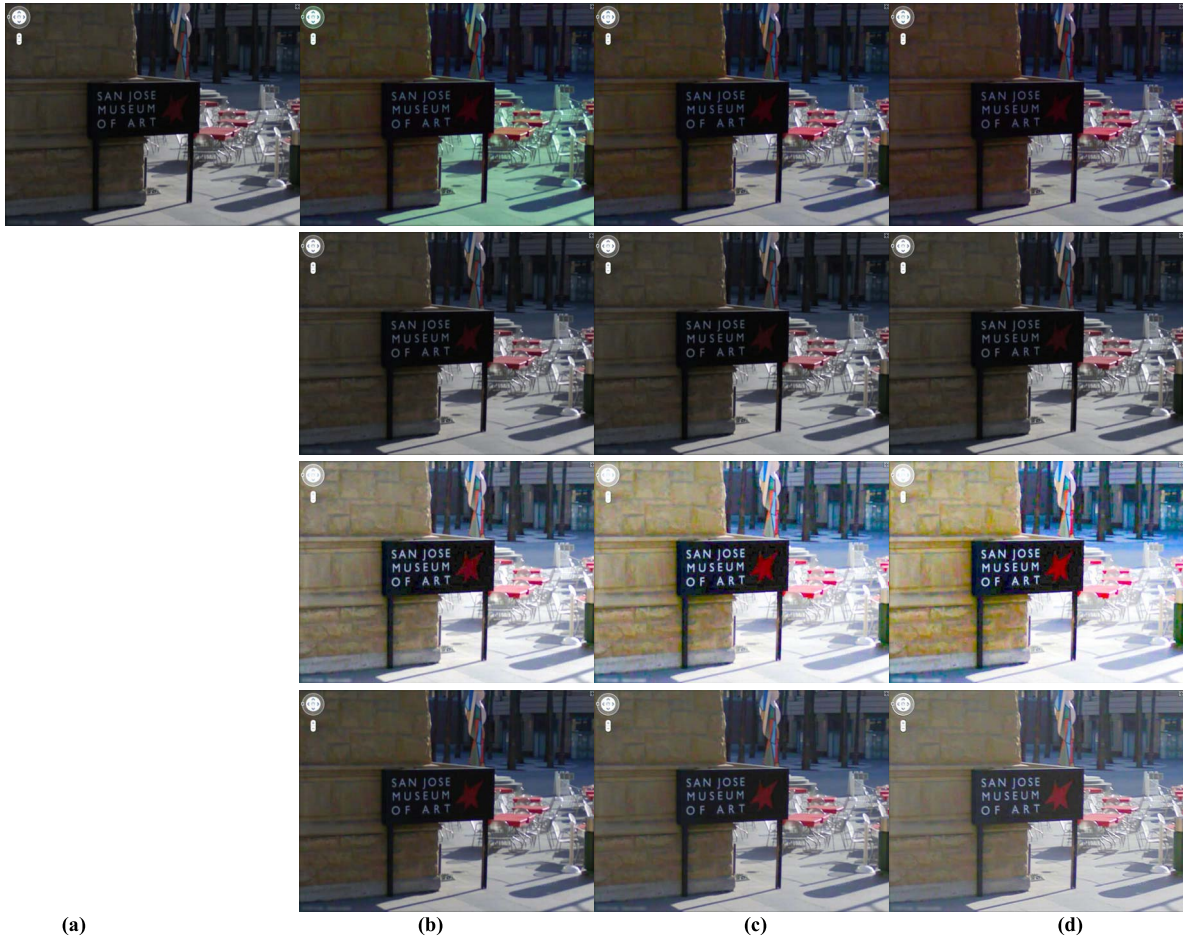


Fig. 17. Experimental results in *Billboard*. (a) Original image. (b)-(d) Its enhanced results under 500lux, 5000lux, and 10000lux. Top to bottom: [12], [16], [14], and the proposed method.

TABLE III
PERFORMANCE COMPARISON OF THE PROPOSED METHOD, [12], [14], [16]

Methods	AMBE (cd/m ²)			DE			EME		
	500lux	5000lux	10000lux	500lux	5000lux	10000lux	500lux	5000lux	10000lux
Original image			0			6.8983			17.3977
[12]	0.3217	0.2903	0.3195	6.8899	6.8802	6.8983	17.5578	17.5543	17.5200
[14]	14.8575	22.0540	25.6769	7.4707	7.4271	7.3422	28.4183	31.2791	34.3638
[16]	0.2241	1.5410	2.6793	6.9230	6.9658	7.0011	16.6406	16.4923	16.4701
Proposed method	3.1110	7.8563	15.4418	7.0147	7.1120	7.1128	17.7117	16.3438	15.0879

performance in AMBE, DE, and EME. It can be observed that [14] achieves the best performance in AMBE, DE, and EME. The proposed method also achieves a good performance in the three metrics. That is, [14] and our method effectively achieve good readability of displayed images under ambient light in terms of luminance and contrast enhancement.

2) *Evaluation of perceptual image distortions*: To evaluate perceptual distortions of displayed images compared with the original images, we utilize two more metrics of dynamic range independent metric (DRI) [38] and visual difference predictor for high dynamic range images (HDR-VDP 2.2) [28]. DRI [38] indicates image distortions in terms of loss of visible contrast, amplification of invisible contrast, and reversal of visible contrast [38].

HDR-VDP 2.2 [28] measures overall image quality of the enhanced images in terms of visual difference compared with their original images. Both metrics mimic HVS perception process, which are effective in evaluating perceptual distortions of displayed images under ambient light. In the DRI evaluation, we do not get amplification of invisible contrast because amplification of invisible contrast improves the readability of displayed images in the dark region under ambient light. As shown in Fig. 16, green pixels represent loss of visible contrast distortions, while red ones represent reversal of visible contrast. More saturated colors in an image indicate more serious distortions. It can be observed that [12] and [16] achieves a good performance but is not effective in luminance and contrast enhancement (TABLE III). The proposed

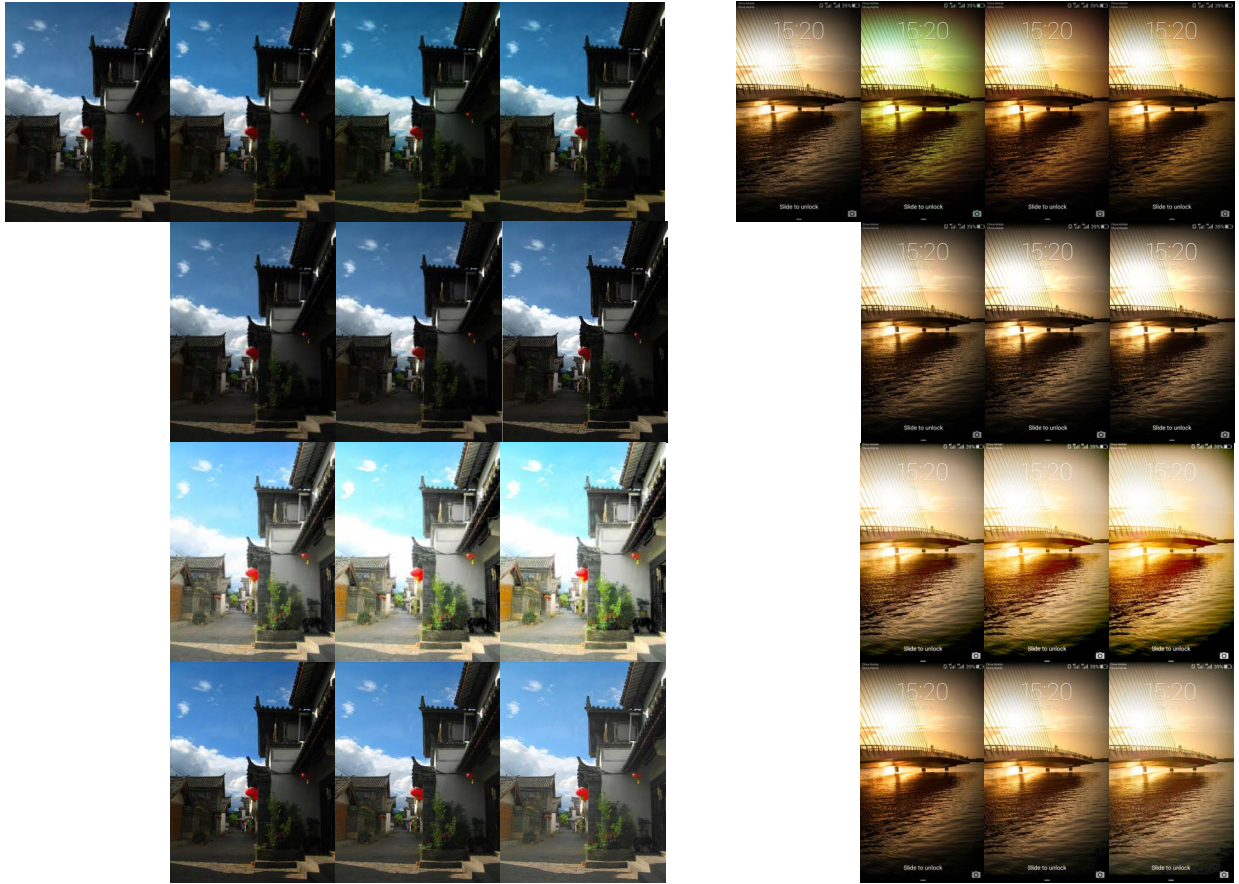


Fig. 18. Experimental results in *Alley* and *Mobile Screen 1*. Left to right: Original image and its enhanced results under 500lux, 5000lux, and 10000lux. Top to bottom: [12], [16], [14], and the proposed method.

TABLE IV
HDR-VDP 2 [28] ASSESSMENT OF THE
PROPOSED METHOD, [12], [14], [16]

Methods	500lux	5000lux	10000lux
Original image	100	100	100
[12]	75.3249	75.7077	74.6044
[14]	48.7415	44.2054	40.0027
[16]	76.8389	65.9000	63.3239
Proposed method	57.5905	53.1504	49.7641

TABLE V
NIQMC [41] ASSESSMENT OF THE PROPOSED
METHOD, [12], [14], [16]

Methods	500lux	5000lux	10000lux
Original image			4.7465
[12]	4.4766	4.7311	4.7328
[14]	5.5942	5.5856	5.5325
[16]	4.7765	4.8175	4.8599
Proposed method	4.8897	4.9972	4.9857

Bold numbers represent the best performance.

method achieves less distortion in readability enhancement than [14].

TABLE IV shows average HDR-VDP 2 performance of four methods. If HDR-VDP 2 is 100, it indicates the best quality (the lower values, the lower quality). In fact, image enhancement causes visual difference which is beneficial for readability enhancement under ambient light to some extent. Therefore, the proposed method and [14] achieve somewhat worse performance in HDR-VDP 2 than [12] and [16].

Among the first and second part in the objective evaluation, the proposed method enhances readability while causing the least distortion under ambient light. The results are consistent with subjective evaluations. This is because the proposed method increases luminance in displayed images without producing distortions such as clipping in [14] and applies gradient enhancement to compensate for gradient distortion.

3) *Evaluation of perceptual visual quality*: We perform perceptual visual quality assessments in terms of three visual

quality indices: No reference (NR) image quality metric for contrast distortion (NIQMC) [41], NR Free Energy Based Robust Metric (NFERM) [42], and blind quality measure for Screen content image (SCI) (BQMS) [43]. NIQMC [41] provides blind image quality assessment for contrast change by combining local and global consideration. The higher the NIQMC [39] value is, the more information and uniform histogram the image contains. TABLE V shows the average NIQMC scores of the proposed method, [12], [14], [16]. It can be observed that [14] and the proposed method achieve better performance than the others. The strong contrast enhancement in [14] flattens the histogram, which provides better local entropy (information) as listed in TABLE III and more uniform histogram in the enhanced images. However, NIQMC [41] is not effective in locally evaluating over-enhancement of contrast (see the third rows of Figs. 15, 17, and 19) in terms

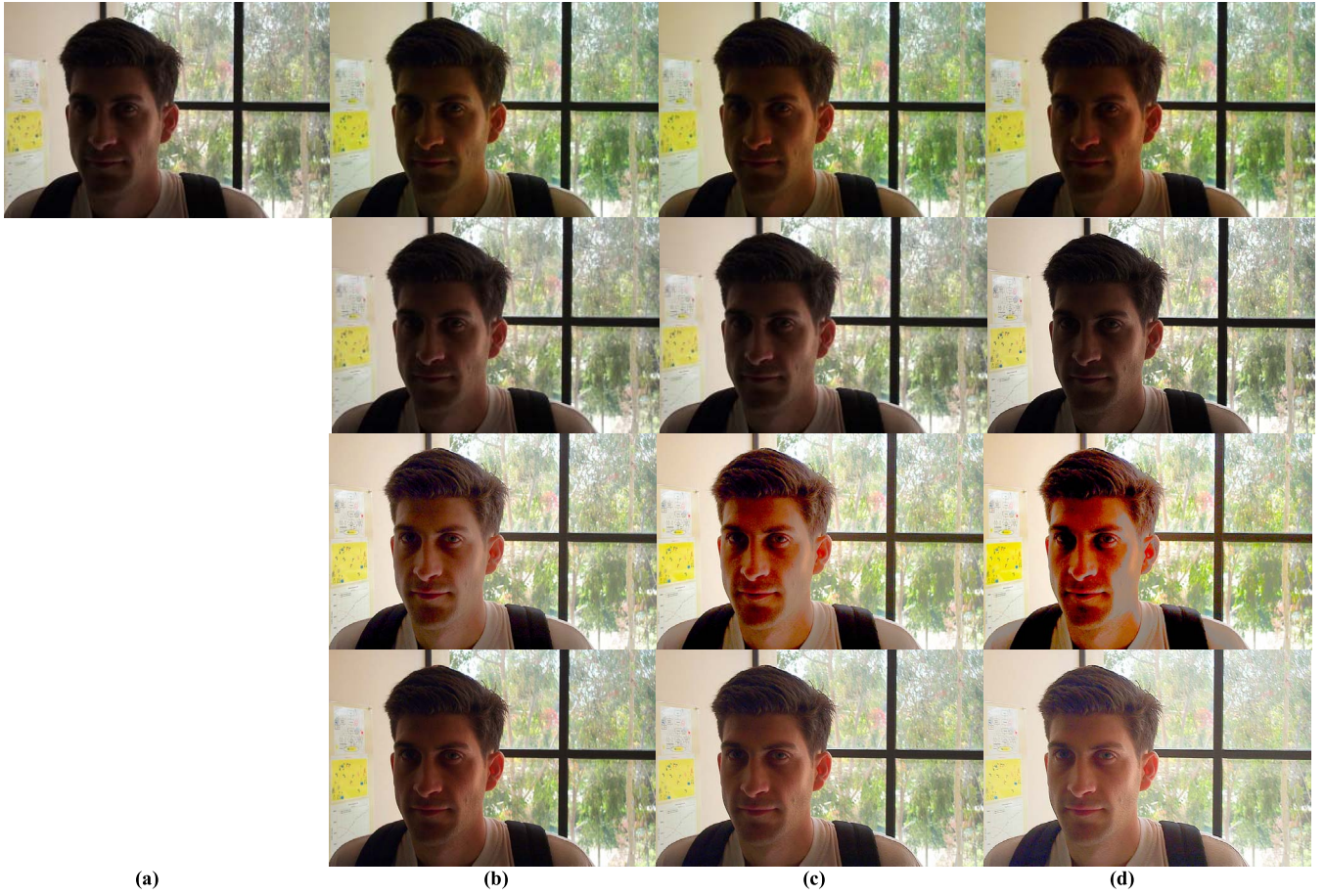


Fig. 19. Experimental results in *Guy*. (a) Original image. (b)-(d) Its enhanced results under 500lux, 5000lux, and 10000lux. Top to bottom: [12], [16], [14], and the proposed method.

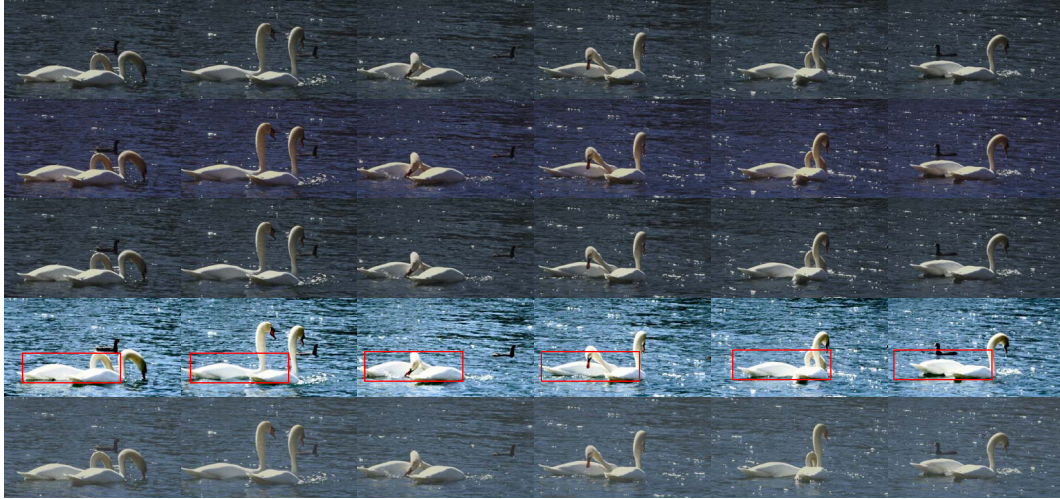


Fig. 20. Six frames in *Swan* sequence and their enhancement results by four methods under 5000 lux. Top to bottom: Original frames, [12], [16], [14], and the proposed method.

of global histogram from part (local considerations) or whole image (global considerations). Thus, we provide more evaluations on the four methods in natural scene images and screen content images in terms of NFERM [42] and BQMS [43]. NFERM evaluates visual quality of natural scenes by extracting the features from free energy, the structural

degradation information, Human Visual System (HVS) inspired features, and Natural Scene Statistics (NSS) model. BQMS assesses visual quality of screen content images from 13 perceptual-inspired features with the free energy and structural degradation model. The higher NFERM [42] and BQMS [43] values are, the better visual quality the image

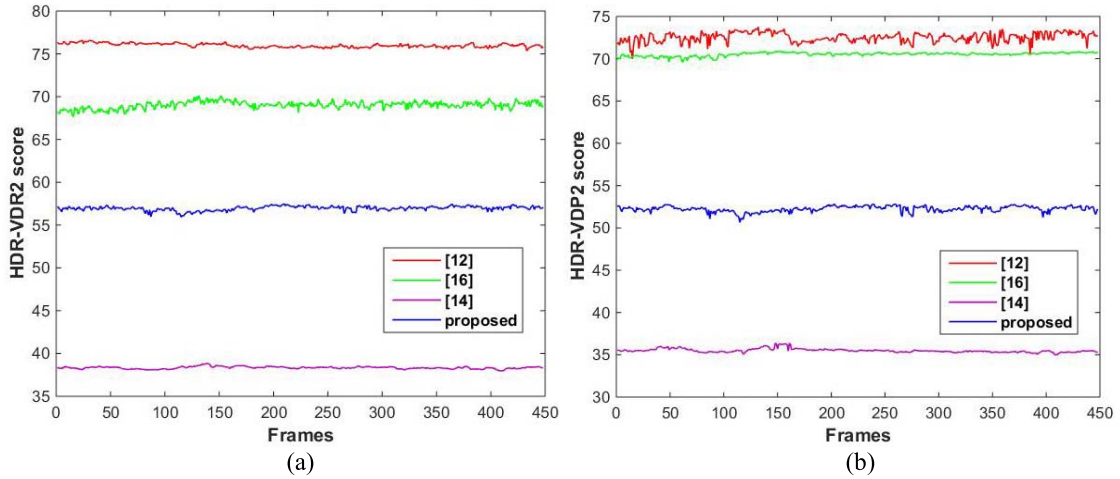


Fig. 21. HDR-VDP2 assessment in *Pool* sequence with different methods under (a) 5000lux and (b) 10000lux.

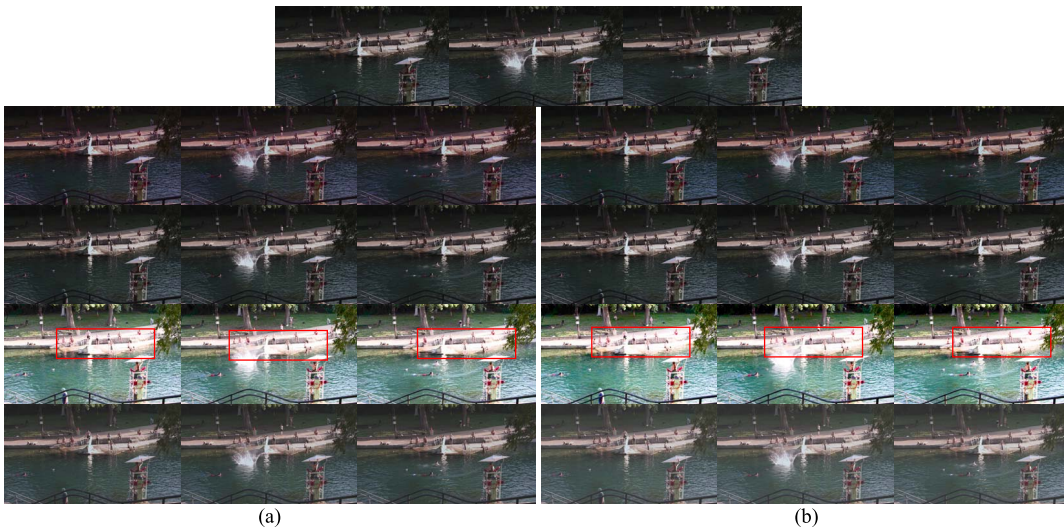


Fig. 22. Three frames in *Pool* sequence and their enhancement results by four methods under (a) 5000lux and (b) 10000lux. Top to bottom: Original frames, [12], [16], [14], and the proposed method.

is. Table VI shows the average scores of NFERM [42] and BQMS [43]. The proposed method achieves the best performance in BQMS on screen content images. In NFERM [16] and [42] produces the best visual quality in 5000lux and 10000lux, but doesn't achieve good luminance enhancement as shown in Table III. The proposed method achieves the best visual quality in 500lux and somewhat worse ones in 5000lux and 10000lux than [16] because of strong luminance and detail enhancement with the increase of ambient light.

C. Video Enhancement

Moreover, we perform video enhancement on two video sequences (*Swan* and *Pool*) from LIVE mobile video quality dataset [44]. *Swan* and *Pool* sequences have a resolution of 1280×720 with 450 and 448 frames, respectively. Figs. 20 and 22 show the enhancement results of six samples in *Swan* and three samples in *Pool*, respectively. [12] causes the color cast problem, while [16] doesn't provide good luminance enhancement as shown in TABLE III. The proposed method enhances the readability of video sequences while achieving

TABLE VI
NFERM [42] AND BQMS [43] ASSESSMENT OF THE PROPOSED METHOD, [12], [14], [16] FOR NATURAL SCENE AND SCREEN CONTENT IMAGES

Methods	500lux		5000lux		10000lux	
	Natural	Screen	Natural	Screen	Natural	Screen
Original image	24.2887	0.8215	24.2887	0.8215	24.2887	0.8215
[12]	25.3001	0.8181	24.4052	0.8178	24.3874	0.8313
[14]	19.8718	0.7993	18.9860	0.7937	19.3145	0.7930
[16]	26.7456	0.7895	25.8348	0.8133	25.0803	0.8146
Proposed	28.7190	0.8209	24.8291	0.8203	22.7769	0.8313

Bold numbers represent the best performance.

less distortion than [14] which causes over-enhancement in contrast (See the red boxes). Moreover, to evaluate perceptual distortions in the enhanced videos, we apply HDR-VDP2 metric [28] to the enhancement results by four methods. As shown in Fig. 21, the proposed method consistently achieves less perceptual distortion than [14] on all frames.

IV. CONCLUSION

In this paper, we have proposed the weighted optimization framework to enhance the readability of displayed images

while causing the least distortion under ambient light. We have provided an ambient light adaptive nonlinear transfer function to compensate for luminance distortions. Also, we have applied TVIA function to achieve detail enhancement under ambient light. To compensate distortions caused by luminance enhancement, we have used the luminance of the enhanced image as weights for more detail enhancement in the bright region than in the dark one. To solve the weighted optimization framework, we have used a numerical solver. Experimental results demonstrate that the proposed method remarkably enhances the readability of displayed images under ambient light by improving details while enhancing luminance. In our future work, we will investigate the effect of ambient light on display color gamut and specular reflection. Also, we perform display enhancement under complicated ambient light.

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